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INCH-POUND

MIL-STD-1698
1 May 1978

DEPARTMENT OF DEFENSE

INTERFACE STANDARD

INSERT ARRANGEMENTS FOR MIL-C-28840(EC) HIGH DENSITY HIGH SHOCK CIRCULAR ELECTRICAL CONNECTORS



AMSC N/A

FSC 5935

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DEPARTMENT OF DEFENSE
WASHINGTON, D.C. 20301

Insert Arrangements for MIL-C-28840(EC) (Classes D, DS, DJ and DJS)
High Density, High Shock, Circular, Electrical Connectors

MIL-STD-1698(EC)

1. This Military Standard is approved for use by the Naval Electronics Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, Department of the Navy, Washington DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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TO ALL HOLDERS OF MIL-STD-1698(EC)

1. THE FOLLOWING PAGES OF MIL-STD-1698(EC) HAVE BEEN REVISED AND SUPERSEDE THE PAGES LISTED:

NEW PAGE	DATE	SUPERSEDED PAGE	DATE
3	1 May 1978	(REPRINTED WITHOUT CHANGE)	
iii	26 May 1981	iii	1 May 1978
4	26 May 1981	4	1 May 1978
90.1	26 May 1981		

2. RETAIN THIS NOTICE AND INSERT BEFORE TABLE OF CONTENTS.

3. Holders of MIL-STD-1698(EC) will verify that page changes and additions indicated above have been entered. This notice page will be retained as a check sheet. This issuance, together with appended pages, is a separate publication. Each notice is to be retained by stocking points until the Military Standard is completely revised or canceled.

Review activities:
Navy - SH

User activities:
Navy - OS

Preparing activity:
Navy - EC
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MILITARY STANDARD

INSERT ARRANGEMENTS FOR

HIGH SHOCK CIRCULAR ELECTRICAL CONNECTORS

TO ALL HOLDERS OF MIL-STD-1698 (EC)

1. THE FOLLOWING PAGE OF MIL-STD-1698(EC) HAS BEEN REVISED AND SUPERSEDES THE PAGE LISTED

NEW PAGE	DATE	SUPERSEDED PAGE	DATE
10.1	16 March 1979	10.1	1 May 1978

2. RETAIN THIS NOTICE AND INSERT BEFORE TABLE OF CONTENTS.

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Supersedes page iii of 1 May 1978

1. SCOPE

1.1 This standard covers insert arrangements for use with MIL-C-28840 (classes D, DS, DJ and DJS) electrical, circular connectors.

2. REFERENCED DOCUMENTS

2.1 Issues of documents. The following document of the issue in effect on date of invitation for bids or request for proposal, forms a part of this standard to the extent specified herein.

SPECIFICATION

MILITARY

MIL-C-28840 - Connectors, Electrical, Circular Threaded, High Shock, High Density, Shipboard, Class D General Specification for

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. DEFINITIONS

3.1 The terms used in this standard are generally accepted by the electrical and electronics industries and commonly used in electrical connector engineering practice.

4. GENERAL REQUIREMENTS

Not applicable.

5. DETAIL REQUIREMENTS

5.1 Dimensions. Dimensions shall be in accordance with the applicable section of this standard and the following dimensional data:

(a) Dimensioning and tolerancing in accordance with ANSI Y14.5-1973. Dimensions are true position and are in inches.
(b) Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
(c) Dimensions and markings shown are for engaging face of pin insert. Socket insert is opposite.
(d) The following tolerances apply to insert installed in shell:
1. The center of each hole in insert for crimp contact connectors shall be located at true position within .010 dia. Ⓜ .010 dia.

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2. The center of engaging end of each contact shall be located in true position within .018 dia. [⌖ .018 dia.]

(e) Unless otherwise indicated, dimensions are symmetrical about centerline. ↑ indicates ℄ of insert arrangement.

(f) Each insert arrangement is shown in the "normal position" in the shell.

(g) Polarization shall be in accordance with MIL-C-28840.

5.2 Main key or keyway polarization (see figure 1)

(a) Each insert arrangement is shown in the "normal position" in the shell.

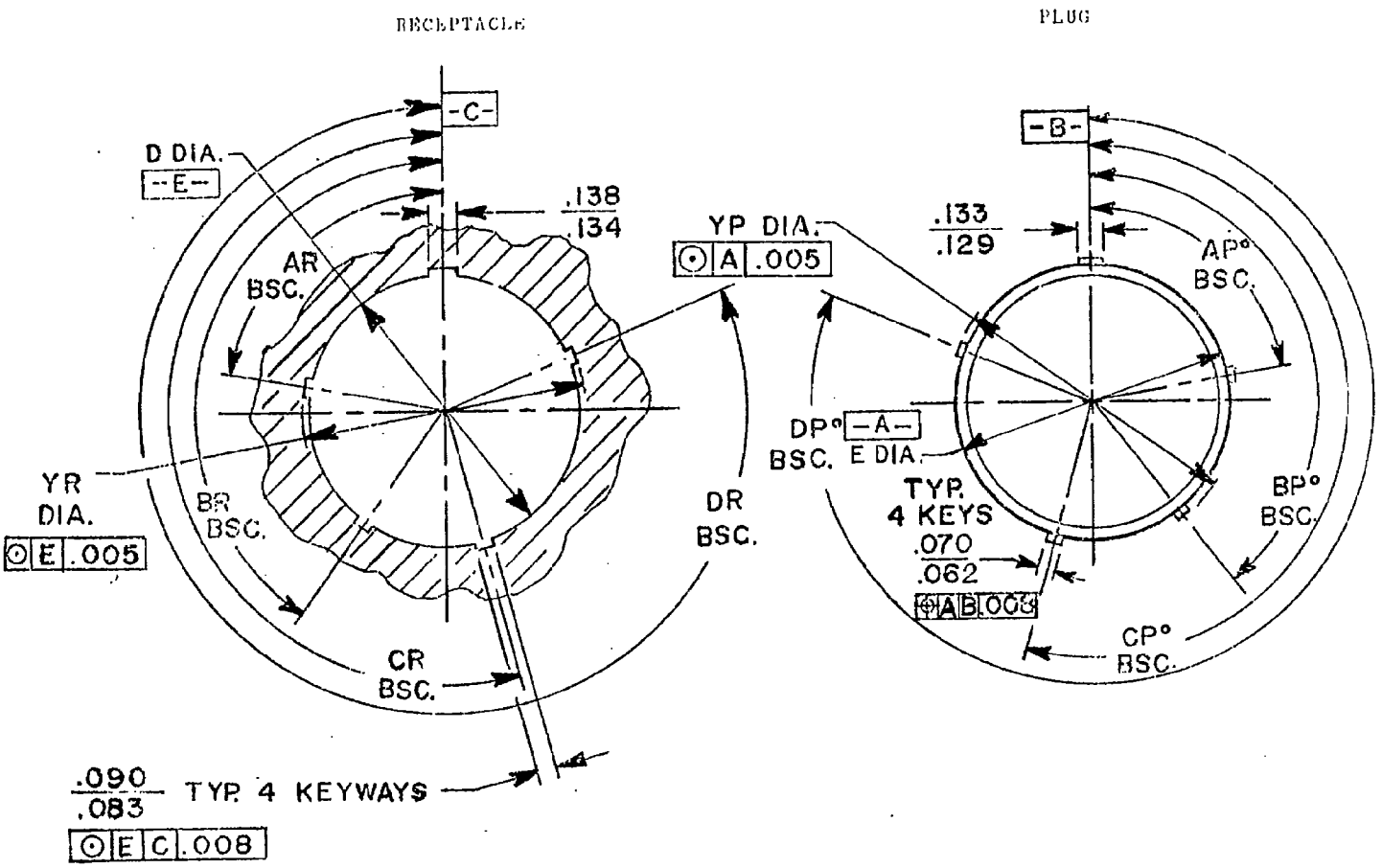
(b) In "alternate keying position" the keys or keyways are positioned as indicated with reference to the master key or keyway.

5.3 Contacts. Contacts shall be in accordance with MIL-C-39029/83 and MIL-C-39029/84.

5.4 Marking. Marking shall be in accordance with MIL-C-28840 and as shown in the applicable section of this military standard.

Preparing activity:
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INCHES	MM
.005	0.13
.008	0.20
.062	1.57
.070	1.78
.083	2.11
.090	2.29
.129	3.28
.133	3.38
.134	3.40
.138	3.50

Figure 1. Connector, electrical, position key and keyways, mating.

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DESIGNATOR 1/	KEY & KEYWAY ARRANGEMENT	AR° or AP° BSC	BR° or BP° BSC	CR° or CP° BSC	DR° or DP° BSC	YP DIA.	YR DIA.
	1	95	141	208	236	.559 (14.20)	.581 (14.76)
A(11)	2	113	156	182	292	.551 (14.00)	.569 (14.45)
	3	90	145	195	252		
	4	53	156	220	255	.683 (17.35)	.705 (17.91)
B(13)	5	119	146	176	298	.675 (17.14)	.693 (17.60)
	6	51	143	184	242		
	1	80	142	196	293	.855 (21.72)	.877 (22.28)
C(15)	2	135	170	200	310	.847 (21.51)	.865 (21.97)
	3	49	169	200	244		
	4	66	140	200	257	.925 (23.50)	.947 (24.05)
D(17)	5	62	145	180	280	.917 (23.29)	.935 (23.75)
	6	79	153	197	272		
E(19)	1	80	142	196	293	1.092 (27.74)	1.114 (28.30)
						1.084 (27.53)	1.102 (27.99)
F(23)	2	135	170	200	310	1.277 (32.44)	1.299 (32.99)
						1.269 (32.23)	1.287 (32.69)
G(25)	3	49	169	200	244	1.438 (36.52)	1.460 (37.08)
						1.430 (36.32)	1.448 (36.78)
H(29)	4	66	140	200	257	1.604 (40.74)	1.626 (41.30)
						1.596 (40.54)	1.614 (41.00)
	5	62	145	180	280	1.796 (45.62)	1.818 (46.18)
J(33)	6	79	153	197	272	1.788 (45.42)	1.806 (45.87)

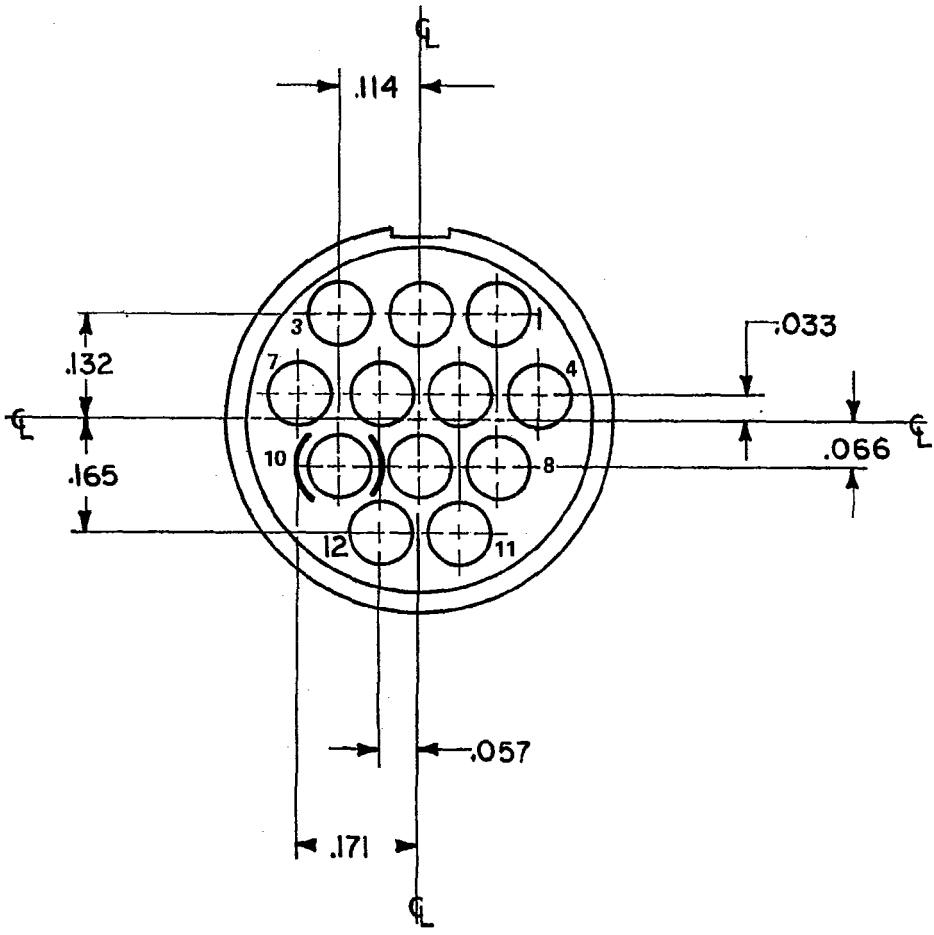
NOTES

1. Shell sizes are provided within parentheses for information and are not a part of the designator.
2. Dimensions are in inches.
3. Metric equivalents (to the Nearest .01 mm) are given for general information only and are based upon 1.0 inch equals 25.4 mm.
4. Dimensions apply after plating.

FIGURE 1. Connector, electrical, position key and keyways, mating.

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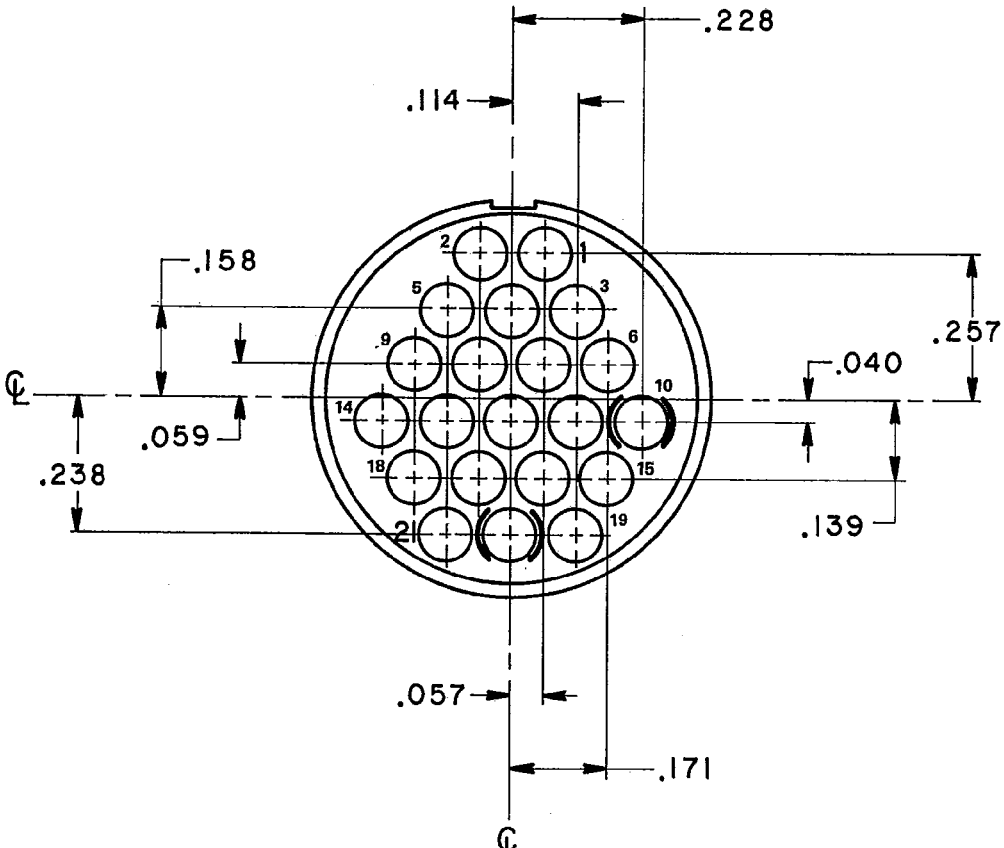
SECTION 10
SHELL SIZE 13



Shell Size Designator	Arrangement Number	Shell Size	Number of Contacts	Size Contacts	Service Rating	Contact Location
B	1	13	12	20	A	All

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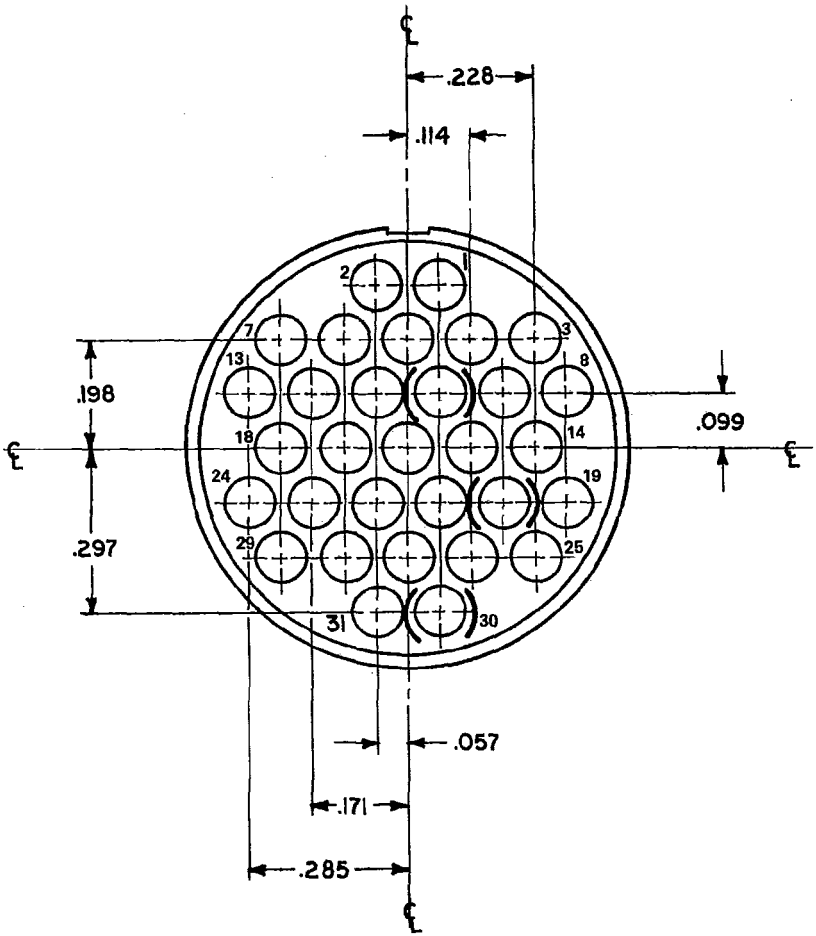
SECTION 20
SHELL SIZE 15



Shell Size Designator	Arrangement Number	Shell Size	Number of Contacts	Size Contacts	Service Rating	Contact Location
C	1	15	21	20	A	A11

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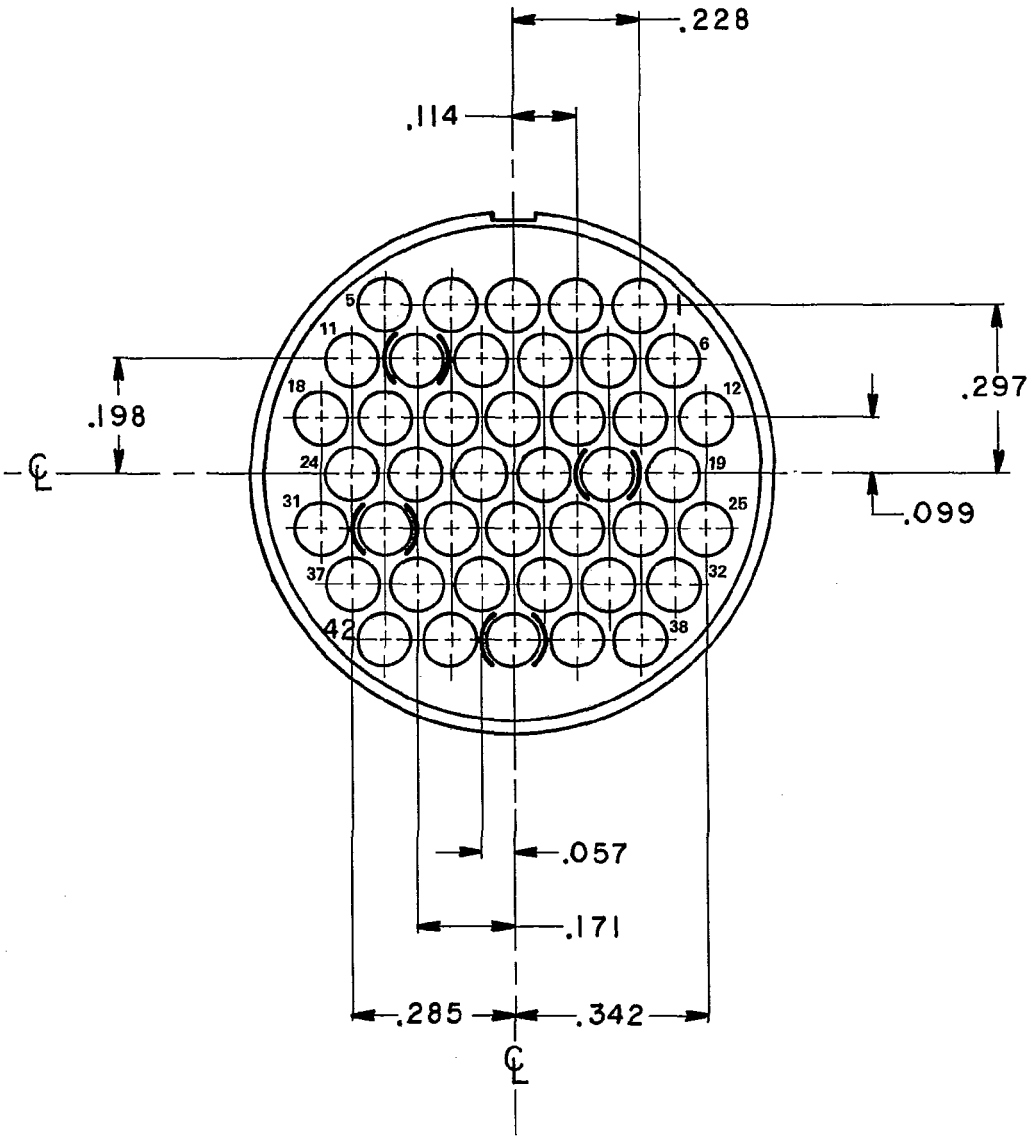
SECTION 30
SHELL SIZE 17



Shell Size Designator	Arrangement Number	Shell Size	Number of Contacts	Size Contacts	Service Rating	Contact Location
D	1	17	31	20	A	All

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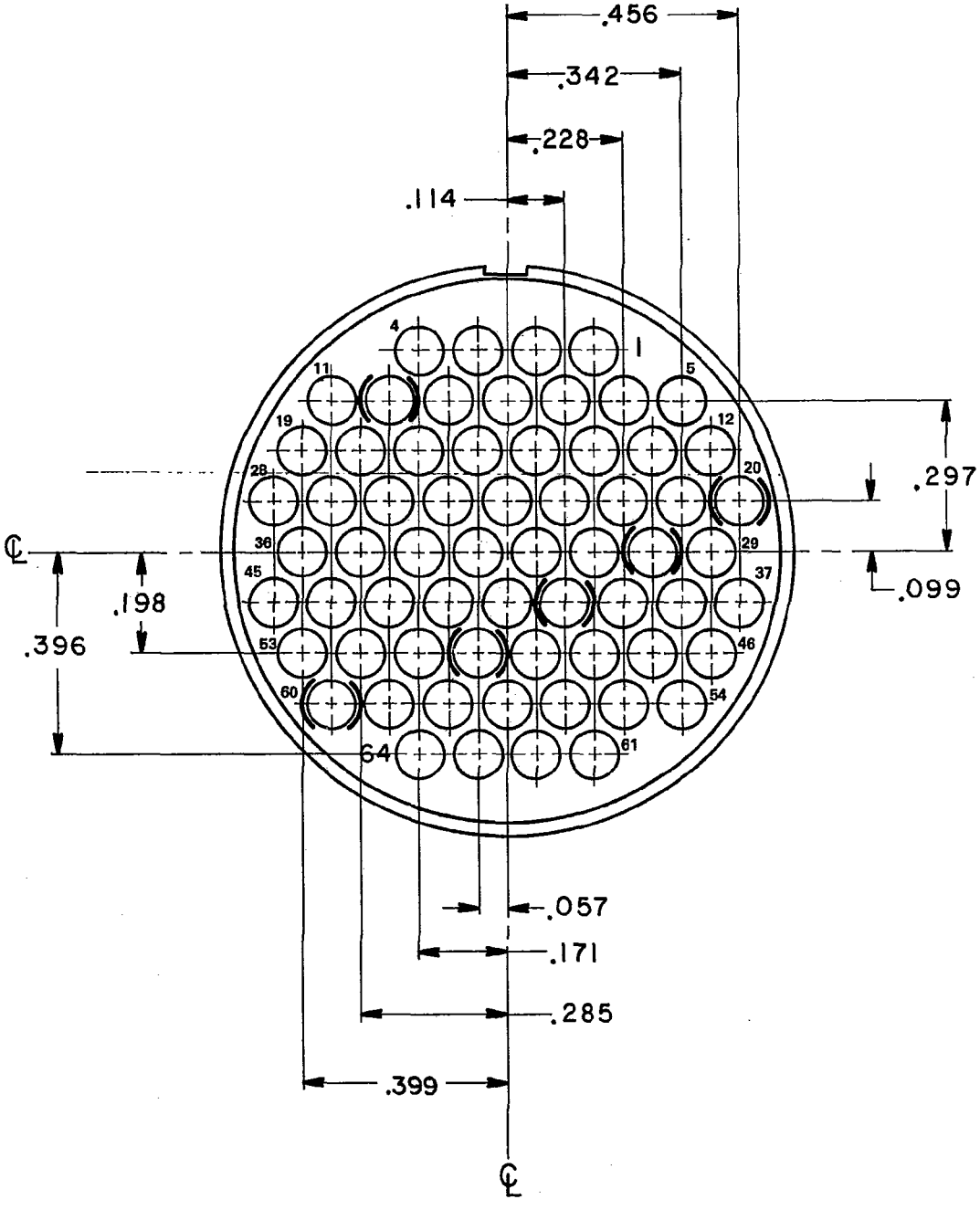
SECTION 40
SHELL SIZE 19



Shell Size Designator	Arrangement Number	Shell Size	Number of Contacts	Size Contacts	Service Rating	Contact Location
E	1	19	42	20	A	A11

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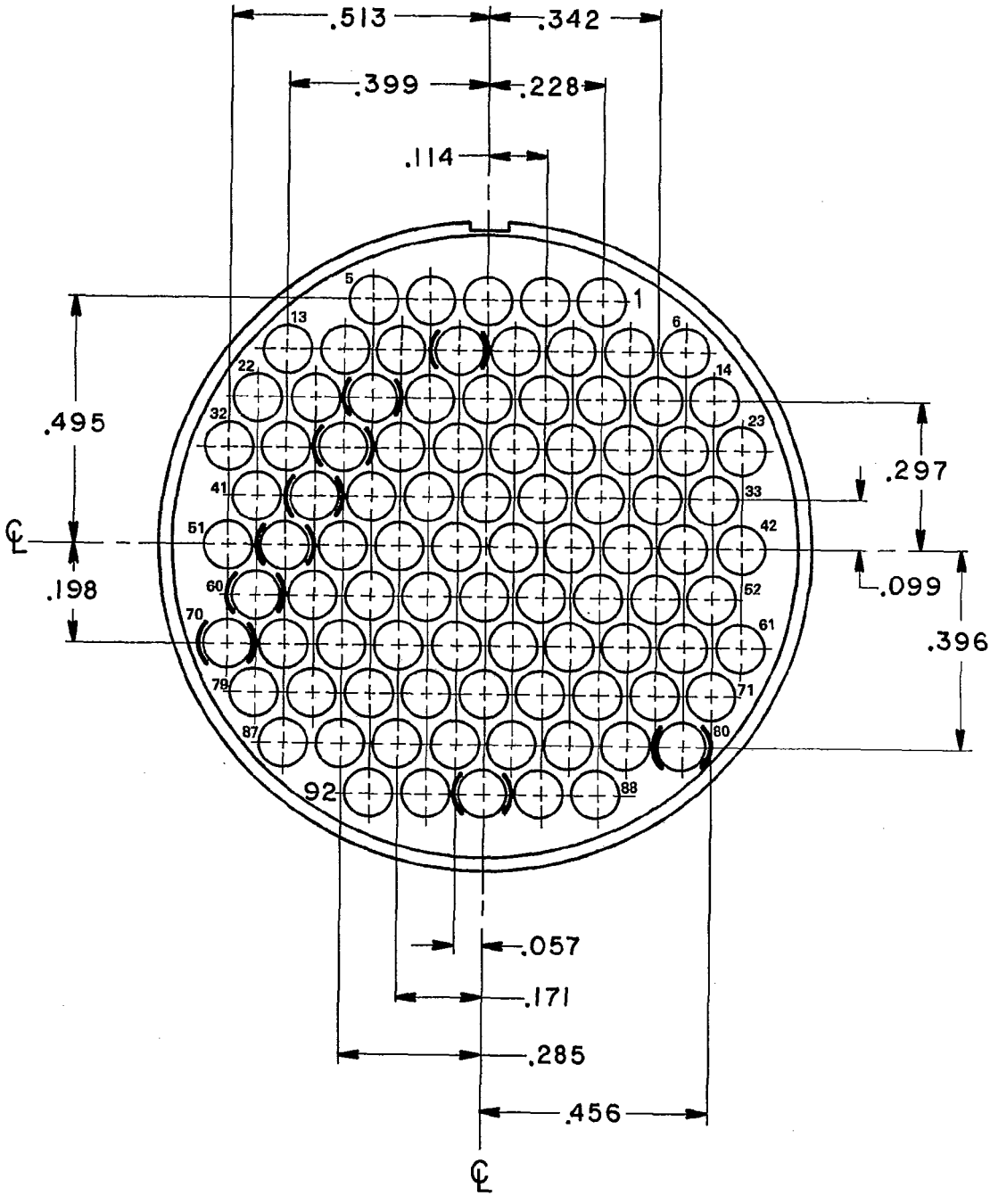
SECTION 50
SHELL SIZE 23



Shell Size Designator	Arrangement Number	Shell Size	Number of Contacts	Size Contacts	Service Rating	Contact Location
F	1	23	64	20	A	A12

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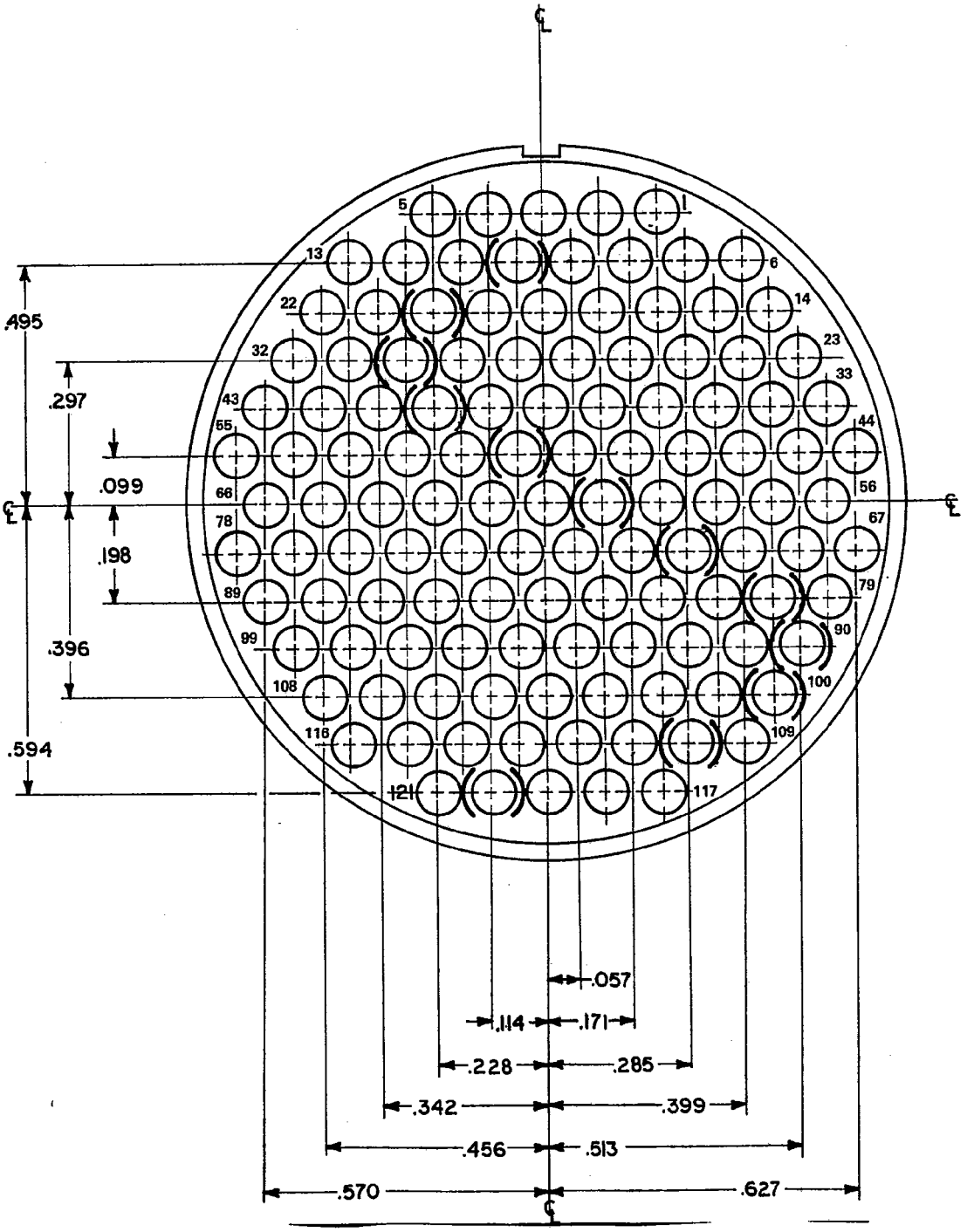
SECTION 60
SHELL SIZE 25



Shell Size Designator	Arrangement Number	Shell Size	Number of Contacts	Size Contacts	Service Rating	Contact Location
G	1	25	92	20	A	A11

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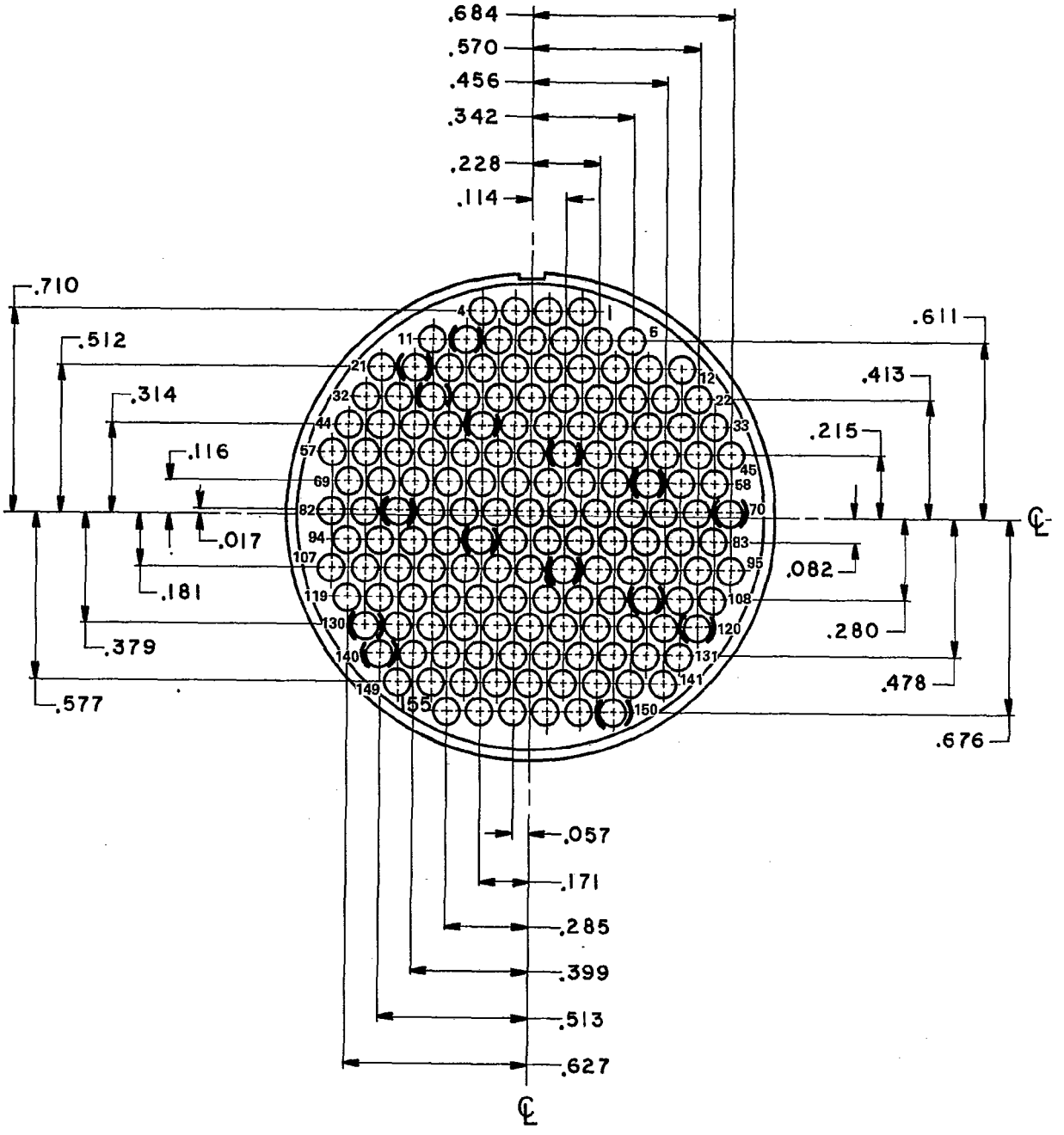
SECTION 70
SHELL SIZE 29



Shell Size Designator	Arrangement Number	Shell Size	Number of Contacts	Size Contacts	Service Rating	Contact Location
H	1	29	121	20	A	All

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SECTION 80
SHELL SIZE 33

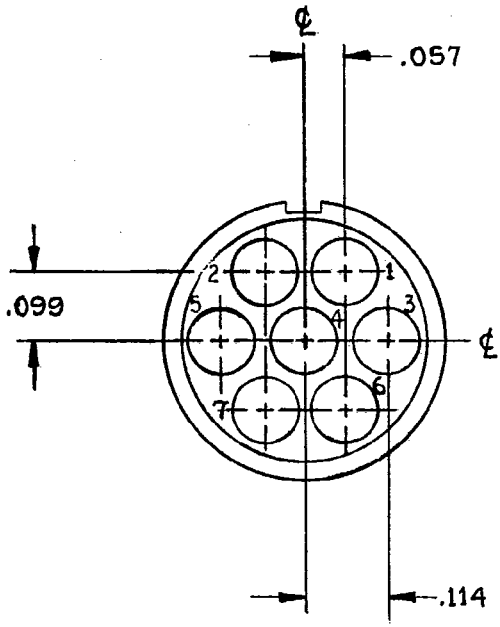


Shell Size Designator	Arrangement Number	Shell Size	Number of Contacts	Size Contacts	Service Rating	Contact Location
J	1	33	155	20	A	All

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SECTION 90

SHELL SIZE 11



SHELL SIZE DESIGNATOR	ARRANGEMENT NUMBER	SHELL SIZE	NUMBER OF CONTACTS	SIZE CONTACT	SERVICE RATING	CONTACT LOCATION
A	1	11	7	20	A	ALL

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